

Skills Summary

Rigid Motions on the Coordinate Plane

Use this reference while completing your worksheet or when studying.

A rigid motion moves the plane without changing any distance. Translations, reflections and rotations are all rigid motions, so an image is always congruent to its original — lengths and angles survive, only position changes.

Skill 1 — Translations

Rule: $(x, y) \rightarrow (x + a, y + b)$. Positive a moves right, negative a moves left; positive b moves up, negative b moves down.

The two coordinates never interact — work on each one separately.

Example: Translate $P(3, -4)$ by $(x, y) \rightarrow (x - 5, y + 2)$.

Step 1. A translation adds a fixed amount to each coordinate on its own, so this is two small sums rather than one combined step.

Step 2. $x' = 3 - 5$ and $y' = -4 + 2$
 $\Rightarrow P'(-2, -2)$

Try it: Translate $(-1, 3)$ by $(x, y) \rightarrow (x + 7, y - 9)$ and give the x -coordinate of the image.

9 :xəp

Skill 2 — Reflections

The five you need: in the x -axis $(x, -y)$; in the y -axis $(-x, y)$; in $y = x$ (y, x) ; in $y = -x$ $(-y, -x)$; in the vertical line $x = c$ $(2c - x, y)$.

Check: the mirror line always sits exactly halfway between a point and its image, so the point and its image are the same distance from it on opposite sides.

Example: Reflect $(12, 7)$ in the vertical line $x = 5$.

Step 1. The mirror is vertical, so only the horizontal position moves — the y -coordinate is untouched and the rule to use is the $x = c$ one.

Step 2. $x' = 2(5) - 12 = 10 - 12$
 $\Rightarrow x' = -2$, and the y -coordinate stays 7.

Try it: Reflect $(10, -6)$ in the line $x = 3$ and give the x -coordinate of the image.

4- :xəp

Skill 3 — Rotations

About the origin: quarter turn counterclockwise $(x, y) \rightarrow (-y, x)$; half turn $(x, y) \rightarrow (-x, -y)$; quarter turn clockwise $(x, y) \rightarrow (y, -x)$.

About any centre (h, k) , quarter turn counterclockwise: $(x, y) \rightarrow (h - (y - k), k + (x - h))$. Measure the point's offset from the centre, turn that offset, and add it back on.

Example: Turn $(4, 6)$ a quarter turn counterclockwise about $(2, 1)$.

Step 1. The centre is not the origin, so the origin rules do not apply directly — use the centre form, which turns the point's offset from the centre.

Step 2. $x' = 2 - (6 - 1) = 2 - 5$ and $y' = 1 + (4 - 2) = 1 + 2$
 $\Rightarrow$ image = $(-3, 3)$

Try it: Turn $(5, 8)$ a quarter turn counterclockwise about the origin and give the x -coordinate.

8- :xpep

(!) **Watch out:** reflecting in $y = x$ swaps the coordinates but changes no signs; reflecting in $y = -x$ swaps them *and* changes both signs. And a rotation about a point other than the origin is never just a sign change — go through the offset form.